

PAPER 1 / QUESTION 135

2 May 2023 / TZ2 (file label) / HL Paper 1 / original question 25 / 2223-6513

- 135.** Wind generator X has a maximum power output P_X for a particular wind speed. For the same wind speed, wind generator Y has a maximum power output P_Y .

The blade radius of Y is three times the blade radius of X. Y is twice as efficient as X.

What is $\frac{P_Y}{P_X}$?

- A. $\frac{3}{2}$
- B. $\frac{9}{2}$
- C. 6
- D. 18